

WORK SHEET - 11

ATOMIC STRUCTURE

Single correct

- Q.1 The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius]
- (A) $\frac{h^2}{4\pi^2 m a_0^2}$ (B) $\frac{h^2}{16\pi^2 m a_0^2}$ (C) $\frac{h^2}{32\pi^2 m a_0^2}$ (D) $\frac{h^2}{64\pi^2 m a_0^2}$
- Q.2 In a hydrogen atom, in transition of electron a photon of energy 2.55 eV is emitted, then the change in wavelength of the electron is
- (A) 3.32 Å (B) 6.64 Å (C) 9.97 Å (D) None of these
- Q.3 A charged particle having specific charge 4×10^{10} col/kg is projected towards nucleus of Zn atom. What is the distance of closest approach if initial speed of projection is 3×10^6 m/s?
- (A) 38.4 pm (B) 3.84 nm (C) 3.84 Å (D) 3.84×10^{-11} nm
- Q.4 In a sample of hydrogen atoms, electrons jump from 10^{th} excited state to ground state. If x are the number of different ultraviolet radiations, y are number of different visible radiations and z are number of different infrared radiations. The value of $z - (x + y)$ is.
[Assume all the Balmer lines lie within visible region]
- (A) 17 (B) 18 (C) 19 (D) 36
- Q.5 If spin quantum number has 4 values instead of two then identify the incorrect statement.
- (A) First period will be shortest with 4 elements
(B) Fourth period can have maximum 36 elements
(C) Each orbital can have maximum 4 electrons.
(D) For a given value of "n" (principal quantum number) number of electrons will be $2n^2$.
- Q.6 The time period of revolution in the 3rd orbit of Li^{2+} ion is x sec. The time period of revolution in the 2nd orbit of He^+ ion, should be
- (A) x sec (B) $\frac{3}{2}$ x sec (C) $\frac{2}{3}$ x sec (D) $\frac{8}{27}$ x sec
- Q.7 In the hydrogen spectrum, the **longest wavelength** in the Lyman series is x Å and the **shortest wavelength** in the Balmer series is y Å. Hence, the **longest wavelength** of light needed to ionise a hydrogen atom from its ground state should be
- (A) $(x + y)$ Å (B) $(y - x)$ Å (C) $\frac{xy}{(x + y)}$ Å (D) $\frac{(x + y)}{xy}$ Å
- Q.8 The orbital angular momentum of an electron is $\sqrt{3} \frac{h}{\pi}$. Which of the following may be the permissible value of angular momentum of this electron revolving in unknown Bohr orbit?
- (A) $\frac{h}{2\pi}$ (B) $\frac{h}{\pi}$ (C) $3 \cdot \frac{h}{2\pi}$ (D) $2 \cdot \frac{h}{\pi}$
- Q.9 A photon of wavelength 12.4 nm is used to emit an electron from the ground state of He^+ . Calculate the debroglie wavelength of emitted electron
- (A) 1.2 Å (B) 2.2 Å (C) 1.8 Å (D) 2.7 Å
- Q.10 When electron jumps from 4th orbit to 2nd orbit in He^+ ion, the radiation emitting out will fall in
- (A) ultraviolet region (B) visible region (C) infrared region (D) radiowave region

- Q.11 For a **unielelectronic specie**, compare the energy of the following orbitals (represented by S)
 S_1 = a spherically symmetrical orbital having two spherical nodes.
 S_2 = an orbital which is double - dumb bell and has no radial nodes.
 S_3 = an orbital with orbital angular momentum zero and three radial nodes.
 S_4 = an orbital having one planar and one radial node.
 (A) $S_1 = S_2 = S_3 = S_4$ (B) $S_1 = S_2 = S_4 < S_3$
 (C) $S_1 > S_2 > S_3 > S_4$ (D) $S_1 < S_2 < S_3 < S_4$
- Q.12 Assuming Heisenberg Uncertainty Principle to be true what could be the minimum uncertainty in de-broglie wavelength of a moving electron accelerated by Potential Difference of 6 V whose uncertainty in position is $\frac{7}{22}$ n.m.
 (A) 6.25 Å (B) 6 Å (C) 0.625 Å (D) 0.3125 Å
- Q.13 When an excited hydrogen atom returned to its ground state, some visible quanta were observed along with other quanta. Which of the following transitions must have occurred?
 (A) $2 \rightarrow 1$ (B) $3 \rightarrow 1$ (C) $3 \rightarrow 2$ (D) $4 \rightarrow 1$
- Q.14 IE_1 for He atom is 24.6 eV. The energy required to remove both the electrons from the atom in its ground state will be
 (A) 59 eV (B) 81 eV (C) 79 eV (D) None of these

More than one correct:

- Q.15 Select the correct statement(s): (According to Bohr model)

(A) Total energy = $\frac{-2\pi^2 Z^2 e^4 m}{(4\pi\epsilon_0)^2 n^2 h^2}$

(B) Radius of orbit = $\frac{(4\pi\epsilon_0)n^2 h^2}{4\pi^2 Ze^2 m}$

(C) Velocity of electron in an orbit = $\frac{2\pi Ze^2}{(4\pi\epsilon_0)nh}$

(D) Frequency of revolution of an electron in an orbit = $\frac{(4\pi\epsilon_0)^2 n^3 h^3}{4\pi Z^2 e^4 m}$

- Q.16 Which of the following options consist of only those orbitals which have number of radial nodes exactly same as their angular nodes?
 (A) 1s, 2p, 3d (B) 1s, 2s, 3s (C) 1s, 3p, 5d (D) 4f, 5g, 6h
- Q.17 In a sample of hydrogen atoms, all atoms are in n^{th} excited state. When they de-excite to the ground state, they emit the radiations of 6 different energies. Which of the following may be characteristic of any of these emitted radiations?
 (A) The wavenumber of atleast one of the radiation can be expressed as sum of wavenumber of one radiation of Paschen and one radiation of Lyman series.
 (B) The frequency of atleast one of the radiation is sum of frequencies of one radiation of Balmer and one radiation of Lyman series.
 (C) The wavelength of atleast one of the radiation is sum of wavelengths of one radiation each from Paschen, Balmer and Lyman series.
 (D) The ratio of longest wavelength to shortest wavelength among the six different energies obtained is 135:7.

- Q.18 Choose the correct statement among the following
- (A) Radial distribution function ($\Psi^2 \cdot 4\pi r^2 dr$) give probability at a particular distance along one chosen direction
- (B) $\Psi^2(r)$ give probability density at a particular distance over a spherical surface
- (C) For 's' orbitals $\Psi(r)\Psi(\theta)\Psi(\phi) = \Psi(x, y, z)$ is independent of θ and ϕ
- (D) '2p' orbital with quantum numbers. $n = 2, \ell = 1, m = 0$, also shows angular dependence
- Q.19 Which of the following statements regarding Bohr's Model of an atom are **incorrect**?
- (A) Energy difference between successive energy levels will keep on decreasing as principal quantum number increases.
- (B) Distance between Bohr's orbit will keep on decreasing as principal quantum number increases.
- (C) Ionisation energy of Li^{+1} is equal to 9 Rydbergs.
- (D) A photon of energy 30eV can cause excitation in He^+ ion present in ground state.

Assertion Reason

- Q.20 **Statement-1:** Energy emitted when an electron jump from $5 \rightarrow 2$ (energy level) is less than when an electron jump from $2 \rightarrow 1$ in all 'H' like atom.
- Statement-2:** The [total energy difference] between 1st & 2nd energy level is greater than that of any two energy level provided level '1' is not part of those two energy levels.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Comprehension :

Paragraph for question nos. 21 to 23

An unknown orbital is having the following equation for its wavefunction.

$$\Psi(r, \theta, \phi) = \frac{2}{3} \left(\frac{Z}{3a_0} \right)^{3/2} (\sigma - 1)(\sigma^2 - 8\sigma + 12) \sigma e^{-\sigma/2} \cos \theta$$

where $\sigma = \frac{2Zr}{a_0}$, a_0 is constant = 0.529 Å and all other parameters have usual meaning. Based on this

data answer the question that follow.

- Q.21 What is the number of total nodes of the orbital? (Total nodes = radial nodes + angular nodes)
- (A) 3 (B) 4 (C) 2 (D) 5
- Q.22 If ' θ ' is the angle measured from z-axis, then the above wavefunction represents which orbital.
- (A) s-orbital (B) d_{xy} orbital (C) p_z orbital (D) d_{z^2} orbital
- Q.23 The maximum distance from the nucleus, where there will a radial node will be given by :
- (A) $r = \frac{a_0}{Z}$ (B) $r = \frac{3a_0}{Z}$ (C) $r = \frac{6a_0}{Z}$ (D) $r = \frac{2a_0}{Z}$

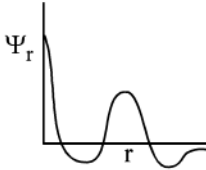
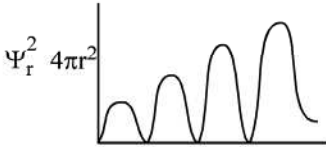
Match the column :

- Q.24 Match the entries in **Column I** with the correctly related quantum number(s) in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	Column II
(A) Orbital angular momentum of the electron in a hydrogen-like atomic orbital	(P) Principal quantum number
(B) A hydrogen-like one-electron wave function obeying Pauli principle	(Q) Azimuthal quantum number
(C) Shape, size and orientation of hydrogen like atomic orbitals	(R) Magnetic quantum number
(D) Probability density of electron at the nucleus in hydrogen-like atom	(S) Electron spin quantum number



- Q.25 Column I & column II contain data on Schrodinger Wave-Mechanical model, where symbols have their usual meanings. Match the columns.

Column I	Column II (Type of orbital)
(A) 	(P) 4s
(B) 	(Q) any 5p orbital
(C) $\Psi(\theta, \phi) = K$ (independent of θ & ϕ)	(R) 3s
(D) at least one angular node is present	(S) $6d_{xy}$

Subjective :

- Q.26 Find the product of multiplication of number of electrons corresponding to $(n + l) = 3$ and $l = 2$ in Cu (29).

- Q.27 Electrons in a sample of H-atoms make transitions from state $n = X$ to some lower excited state. The emission spectrum from the sample is found to contain only the lines belonging to a particular series. If one of the photons had an energy of 0.6375 eV, then find out value of X.

[Take $0.6375 \text{ eV} = \frac{3}{4} \times 0.85 \text{ eV}$]

- Q.28 A sample of hydrogen atoms in ground state absorbs light of $x \text{ nm}$. If the excited sample emits the radiations of six different photon energies but the detector of these emitted radiations has not detected ultraviolet radiations, then the value of 'x' is (Given : $R_H = \frac{1}{96} \times 10^9 \text{ m}^{-1}$)

- Q.29 Two bulbs A and B are emitting monochromatic light of wavelength such that A can just ionise H atoms & B can just ionise He^+ ions. If the power of A & B are 30 W & 40 W respectively. Calculate the ratio of number of photons emitted per second by bulb A to bulb B?

- Q.30 Mr. Santa has to decode a number "ABCDEF" where each alphabet is represented by a single digit. Suppose an orbital whose radial wave function is represented as

$$\Psi_{(r)} = k_1 \cdot e^{-r/k_2} (r^2 - 5k_3r + 6k_3^2)$$

From the following information given about each alphabet then write down the answers in the form of "ABCDEF", for above orbital.

Info A = Value of n where " n " is principal quantum number

Info B = No. of angular nodes

Info C = Azimuthal quantum number of subshell to orbital belongs

Info D = No. of subshells having energy between $(n + 5)s$ to $(n + 5)p$ where n is principal quantum number

Info E = Orbital angular momentum of given orbital.

Info F = Radial distance of the spherical node which is farthest from the nucleus

(Assuming $k_3 = 1$)

Rankers Academy JEE

LIST OF IMPORTANT FORMULAS

<u>Constants</u>	<u>Symbol</u>	<u>SI Value</u>	
Electron rest mass	m_e	$9.10 \times 10^{-31} \text{ kg}$	0.000548 amu
Proton rest mass	m_p	$1.67 \times 10^{-27} \text{ kg}$	1.00757 amu
Neutron rest mass	m_n	$1.67 \times 10^{-27} \text{ kg}$	1.00893 amu
Planck constant	h	$6.62 \times 10^{-34} \text{ J s}$	$6.62 \times 10^{-27} \text{ erg s}$
Bohr magneton (BM)	β_e	$9.27 \times 10^{-24} \text{ J/T}$	
Gas constant	R	8.3145 J/mol-K	0.0821 atm-lit / mol / K

Energy Conversion Factors

$$1 \text{ erg} = 10^{-7} \text{ J}$$

$$1 \text{ cal} = 4.184 \text{ J}$$

$$1 \text{ eV} = 1.602177 \times 10^{-19} \text{ J} = 1.602177 \times 10^{-12} \text{ erg} = 23.0605 \text{ kcal/mol}$$

STRUCTURE OF ATOM

1. Rutherford's Model

$$(a) R_n = R_1 (A)^{1/3}, \quad R_1 = 1.33 \times 10^{-13} \text{ cm} \quad A = \text{mass number} \quad R_n = \text{Radius of nucleus}$$

$$(b) \frac{1}{2} m_\alpha v_\alpha^2 = K \frac{Z \cdot 2e}{r}; \quad r = \text{distance of closest approach}, \quad v_\alpha = \text{initial velocity of a } \alpha\text{-particle}$$

2. Light

(a) Photon is considered massless bundle of energy.

$$(b) E = mc^2$$

$$(c) E_{\text{photon}} = h\nu = hc/\lambda = hc \bar{\nu}$$

$$(d) E = \frac{hc}{\lambda} \cong \frac{1240 \text{ eV} \cdot \text{nm}}{\lambda}$$

3. Bohr's Model

$$(a) \text{ Bohr quantization rule} \quad mvr = n \cdot \frac{h}{2\pi} = n \cdot \hbar$$

$$(b) E_n = \frac{E_1}{n^2} Z^2 = -\frac{2\pi^2 me^4 K^2}{n^2 h^2} Z^2; \quad E_1 = \frac{-2\pi^2 me^4 K^2 Z^2}{h^2}$$

$$(c) r_n = \frac{h^2}{4\pi^2 e^2 m K} \times \frac{n^2}{Z}$$

$$(d) v_n = \frac{2\pi e^2 K}{h} \times \frac{Z}{n}$$

$$(e) \text{ Revolutions per sec} = v/2\pi r$$



4. Spectral lines

- (a) Rydberg's Equation $\frac{1}{\lambda} = \bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \times Z^2$; $R_H \cong 109700 \text{ cm}^{-1}$
- (b) For First line of a series $n_2 = n_1 + 1$ (Also known as α line)
- (c) H_α line means $n_2 = n_1 + 1$; also known as line of longest λ , shortest ν , least E
- (d) Number of spectral line observed in the spectrum = $\frac{n(n-1)}{2}$
when e^- de-excites to ground state, n = number of higher orbit

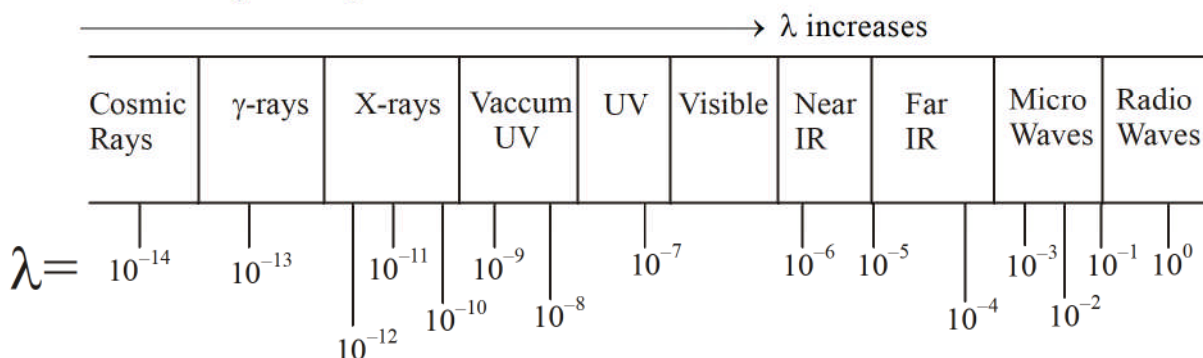
5. Photoelectric effect

- (a) Kinetic energy = $h\nu - w = h\nu - h\nu_0$
where w = work function
 ν_0 = Threshold frequency
- (b) Accelerating potential = $eV = KE = \frac{1}{2}mv^2$
- (c) $\nu^{1/2} = a(z-b)$ b = screening constant

6. De-broglie, Heisenberg & Schrodinger equations

- (a) Number of waves = n = principal quantum number
- (b) $\lambda = h/mv = h/p$
- (c) wavelength of electron (λ) $\cong \sqrt{\frac{150}{V \text{ in volts}}} \text{ \AA}$
- (d) $\Delta x \cdot \Delta p \geq h/4\pi$
- (e) Orbital angular momentum = $\frac{h}{2\pi} \sqrt{l(l+1)}$
- (f) Spin angular momentum = $\frac{h}{2\pi} \sqrt{S(S+1)}$
- (g) Magnetic moment (μ) = $\sqrt{n(n+2)} \text{ B.M.}$; n = number of unpaired electron
- (h) Radial Nodes = $(n - l - 1)$
- (i) Angular nodes = l
- (j) Total nodes = $(n-1)$

7. Electromagnetic Spectrum



ANSWER KEY

Q.1	C	Q.2	B	Q.3	C	Q.4	A	Q.5	D
Q.6	C	Q.7	C	Q.8	D	Q.9	C	Q.10	A
Q.11	B	Q.12	C	Q.13	A	Q.14	C	Q.15	ABC
Q.16	C	Q.17	ABD	Q.18	CD	Q.19	BCD	Q.20	A
Q.21	B	Q.22	C	Q.23	B				
Q.24	(A) Q,R (B) P, Q, R,S (C) P, Q, R (D) P, Q								
Q.25	(A) P, (B) P,Q,S, (C) P, R (D) Q, S					Q.26	80	Q.27	8
Q.28	0100	Q.29	0003	Q.30	300303				

Rankers Academy JEE

